Life Expectancy Changes Due to Kidney Donation Across Race, Sex, and Age

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Abstract
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Methods

Model overview

We constructed a Markov state-transition model to compare lifetime outcomes for two hypothetical cohorts of otherwise identical individuals: one cohort that donates a kidney and one that does not. Both cohorts enter the model at the age of donation in full health, and the simulation advances in annual cycles until age 100. The analysis was conducted from the perspective of the individual donor rather than the health-care system; outcomes are reported as undiscounted remaining life-years, consistent with the standard approach for life-expectancy analyses[1]. The base-case age at donation is 40 years; subgroup analyses span ages 25–55. We used a Monte Carlo simulation with 5×10^6 individuals per arm and cross-validated against a deterministic analytic cohort implementation.

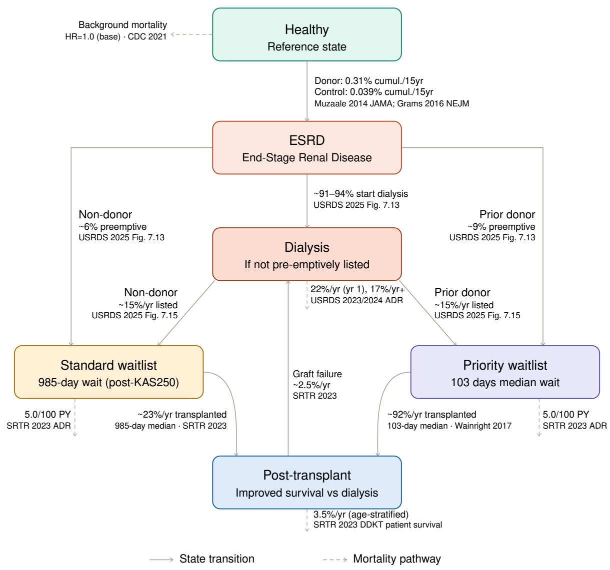

Figure 1. The Markov model for kidney donation.

Health states

The model contains seven mutually exclusive health states (Figure 1): *Healthy*, *ESRD onset*, *Dialysis*, *Standard waitlist*, *Priority waitlist*, *Post-transplant*, and *Dead* (absorbing). At ESRD onset, the cohort branches: a fraction is listed preemptively (bypassing dialysis entirely) and the remainder initiates haemodialysis. The donor arm routes ESRD survivors to the *Priority waitlist*, reflecting the allocation priority afforded to prior living donors (PLD) under KAS250; the non-donor arm routes to the *Standard waitlist*. Graft failure returns patients to the *Dialysis* state, bypassing the ESRD-onset branch.

ESRD risk after donation

Fifteen-year cumulative ESRD incidence was taken from Muzaale et al.[2]: 30.8 per 10,000 (95% CI 24.3–38.5) for donors and 3.9 per 10,000 (95% CI 0.8–8.9) for age-, sex-, and health-matched controls (8-fold relative risk). Race-specific donor rates were 74.7/10,000 (Black) and 22.7/10,000 (White)[2]. The White non-donor baseline was taken from Grams et al.[3] (sex-averaged 0.05%/15 yr), because zero ESRD events were recorded in White matched controls in Muzaale et al.

ESRD hazard was modelled with a Weibull distribution (shape $k = 1.5$), calibrated via a competing-risk bisection algorithm to reproduce the 15-year cumulative incidence while accounting for background-mortality attrition in the at-risk pool. The accelerating hazard ($k > 1$) is supported by the Massie et al. 20-year cumulative incidence of 34 per 10,000 (vs. $\approx$21/10,000 predicted under a constant-hazard model)[4]. Within-donor subgroup hazard ratios were drawn from Massie et al.: 2.96 (95% CI 2.25–3.89) for Black race, 1.88 (95% CI 1.50–2.35) for male sex, and 1.40 per decade (non-Black donors)[4].

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Dialysis survival and waitlist listing

First-year haemodialysis mortality was 22% and subsequent-year mortality 17% ($\approx$42% five-year survival), consistent with USRDS estimates[5].

At ESRD onset, a fraction of patients is listed preemptively before dialysis initiation, avoiding dialysis mortality entirely. Preemptive listing probabilities were drawn from USRDS 2025 ADR Figure 7.13[6]: 5.8% overall for the non-donor arm and 9.4% for the donor arm (age 18–44 cohort, reflecting the younger and healthier donor-like population).

Among patients on dialysis, the conditional annual probability of joining the waitlist was set to 15%/year (base case), derived from USRDS 2025 ADR Figure 7.17[6]. Approximately 25% of 18–44 year-old ESRD patients are listed or transplanted within one year, of whom $\approx$9.4% were preemptively listed, yielding an estimated post-dialysis listing rate of $\approx$15.6%/year for the donor-like cohort (conservative base case: 15%/year; one-way sensitivity range 5–30%/year).

Transplant waiting time and waitlist outcomes

Transplant probability while listed was modelled as a time-homogeneous exponential process parameterised by the median waiting time, a standard approximation for waiting-time analyses[7]. For the standard waitlist, the post-KAS250 national median of 985 days was used[8]; for the PLD priority waitlist, the post-KAS median of 102.6 days was used[9]. Waitlist mortality was 5.0 deaths per 100 patient-years (SRTR 2023 ADR Figure KI 24) and competing removal 6.81%/year (SRTR 2023 ADR Figure KI 22)[10].

Post-transplant survival

Post-transplant annual mortality was derived from age-stratified 5-yearKaplan–Meier estimates for DDKT recipients in the SRTR 2023 ADR (Figure KI 70, 2016–2018 cohort)[10]: 0.9%/year (ages 18–34), 1.8%/year (35–49), 3.9%/year (50–64), and 6.9%/year ($\geq$ 65). The base case uses DDKT survival because prior living donors who develop ESRD predominantly receive transplants via the deceased-donor priority pathway rather than living-donor transplantation, consistent with USRDS data showing that living-donor transplants account for a small minority of transplants in this population[11,12]. A one-way sensitivity analysis substitutes LDKT patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year by age band) to bound the effect of this assumption. Annual graft failure was 2.5%/year post-year 1[10]; graft failure returned patients to the Dialysis state.

Background mortality and donor all-cause survival

Background mortality in the Healthy state was taken from the 2021 US Life Tables (CDC, National Vital Statistics Reports Vol. 72 No. 12)[13], with sex-specific tables used in sex-stratified analyses. The base-case donor all-cause mortality hazard ratio versus matched non-donors is 1.0, consistent with US-based studies[2,14] and the Grams et al. 2018 meta-analysis (pooled HR 0.984, 95% CI 0.743–1.302; 9 studies, n = 118,426 donors)[15]. An HR of 1.30 (Mjøen et al.[16], a Norwegian cohort in which 80% of donors were first-degree relatives of the recipient) was examined as the pessimistic sensitivity scenario.

Subgroup analyses

Age at donation Analyses were repeated at ages 25, 35, 40, 45, and 55 years at donation. Donor ESRD rates were scaled by the Massie 2017 age HR (1.40/decade, non-Black donors)[4]; non-donor baselines were held at the Muzaale overall matched-control rate.

Race Race-specific donor ESRD rates (Black 74.7/10,000; White 22.7/10,000) and race-specific non-donor baselines (Black sex-averaged 0.195%/15 yr; White sex-averaged 0.05%/15 yr) were taken from Muzaale et al.[2] and Grams et al., respectively[3]. Race-specific waitlist mortality and post-transplant survival were from SRTR 2023 Figures KI 25 and KI 71[10].

Sex Sex-specific donor ESRD rates were derived by applying the Massie 2017 male-sex HR (1.88)[4] to the overall donor rate, weighted by the observed donor sex mix (60% female; SRTR 2023 ADR Figure KI 7)[10]. For non-donor baselines, we used the Grams 2016 White sex-specific rates (female 0.04%/15 yr; male 0.06%/15 yr) rather than applying the within-donor sex HR to the control arm, because Massie 2017 hazard ratios describe variation within the donor cohort and are not appropriate for scaling a different reference population[3]. This sourcing rule, Muzaale matched-control rates for overall analyses and Grams direct rates wherever sex or race is stratified, was applied consistently across all matrix subgroup figures.

Probabilistic sensitivity analysis

We conducted a PSA with 200 parameter draws. Sampled quantities and their distributions were: ESRD 15-year cumulative risks (beta distributions parameterised from the Muzaale 2014 published 95% CIs); donor all-cause mortality HR (log-normal, $\mu = -0.016$, $\sigma = 0.143$, derived from the Grams 2018 meta-analysis CI); Weibull shape k (log-normal, $\sigma = 0.13$, representing structural uncertainty absent published CIs for this parameter); and waitlist listing probability (beta, fractional SE = 40%, consistent with the 0.05–0.30 plausible range). Registry-derived parameters (SRTR, USRDS) have negligible sampling error relative to structural uncertainty and were fixed at base-case values; scenario variation in these parameters is captured by the one-way sensitivity analysis. We report the mean ΔLE, 95% credible interval, and the probability that donation is life-expectancy–neutral or beneficial.

One-way sensitivity analysis

Each parameter was varied individually while others were held at base-case values. Key scenarios included: no priority access (PLD wait set to the standard 985-day median), donor all-cause mortality HR = 1.30 (Mjøen pessimistic estimate), ESRD relative risk $\times$11 (Mjøen upper-bound ESRD HR), PLD wait 50 days (optimistic), post-transplant survival at LDKT quality (SRTR Figure KI 76), and dialysis mortality $\pm$50%.

Calibration

The model's transplant probability was validated against the SRTR 2023 3-year waitlist outcome distribution (Figure KI 22, 2019–2021 listing cohort) [10]. This comparison is non-circular because the transplant probability was derived from the median waiting time, not from the 3-year outcome distribution. The model predicted 44.3% transplanted at three years versus 44.3% observed (DDKT + LDKT combined; $\Delta = 0\%$). The waitlist removal rate was back-calculated via a competing-risk bisection algorithm to reproduce the SRTR-observed 19.1% removed at three years (12.6%/year); the naïve formula $1 - (1 - \text{CIF})^{1/3}$, which ignores competing depletion by transplantation and death, was not used. The model over-predicted three-year waitlist mortality (10.0% vs. 6.8% observed, $\Delta = +47\%$), likely reflecting the application of a 2023 cross-sectional mortality rate (confirmed 5.0/100 patient-years from the raw SRTR data file) to a 2019–2021 listing cohort; this bias is symmetric across both arms and conservative (slightly overstates the cost of the waitlist period for both donors and non-donors equally). No independent cohort exists tracing the complete pathway from living donation through ESRD onset to transplantation; these checks represent the available face-validity evidence for the model.

Supplementary analyses: voucher benefit and ESRD-conditional priority benefit

Two supplementary analytic-cohort Markov models were built on the same state structure, transition probabilities, and parameter sources as the base-case donor model, in order to decompose the population-level donor ΔLE result into its constituent mechanisms.

Voucher-holder benefit The first model quantifies the life-expectancy effect of a kidney-donation "voucher": a mechanism by which a living donor designates a healthy, non-donor family member to receive PLD priority-waitlist access should that family member ever develop ESRD. Two arms of healthy non-donors were simulated, both carrying non-donor baseline ESRD risk and a background all-cause mortality HR of 1.0; the sole difference between arms was waitlist priority upon ESRD onset (voucher arm: PLD median wait 102.6 days [9]; control arm: standard median wait 985 days [8]). Consistent with the sourcing rule used for the base-case preemptive-listing parameters, the voucher arm was assigned the informed/monitored preemptive-listing probability (9.4%) and the control arm the general-population probability (5.8%) [6]. All other transition probabilities (dialysis mortality, per-cycle waitlist listing, waitlist mortality and removal, post-transplant survival, graft failure) were identical to the base-case model [10]. The analysis was run at the base-case age of 40, with an age sweep (25–60 years) and an ESRD-risk subgroup analysis using the Muzaale overall and Grams race-specific non-donor CIFs [2,3]. One-way sensitivity analyses varied the voucher and standard median wait times, the per-cycle listing probability, the preemptive-listing probability, and post-transplant graft quality (DDKT vs. LDKT).

ESRD-conditional priority-waitlist benefit Because the voucher (and, by extension, the donor PLD-priority) benefit is diluted at the population level by the large fraction of the cohort who never develop ESRD, a second model conditioned entirely on ESRD onset: all individuals entered the model already diagnosed with ESRD, in order to isolate the life-expectancy benefit of priority waitlist access for the sub-population who actually reach the waitlist. Entry to dialysis versus preemptive listing, dialysis and waitlist mortality, waitlist removal, post-transplant survival, and graft failure used the same age-stratified, all-cause parameters as the base-case model; no separate Healthy state or background-mortality term was included, since dialysis- and waitlist-state mortality rates are all-cause rates already measured in ESRD populations. Two arms were compared, differing only in waitlist priority (priority vs. standard median wait, as above). The base case used an ESRD-onset age of 60, with an age sweep (40–75 years at ESRD onset) and one-way sensitivity analyses on wait times, per-cycle listing probability, preemptive-listing probability, post-transplant graft quality, and dialysis mortality ($\pm 20\%$).

Both supplementary models used a deterministic analytic-cohort implementation (as in the base-case cross-validation) with 10^6 individuals per arm.

Analytic cycle length and discounting

Annual (one-year) cycles were used throughout. Life-years accumulated with an end-of-cycle convention; the resulting systematic underestimate of approximately 0.5 years applies equally to both arms and cancels in ΔLE. Future life-years were not discounted, consistent with the convention for pure life-expectancy analyses [1].

Software and reproducibility

All analyses were implemented in Python 3.10 (NumPy, pandas, matplotlib). The simulation, parameter assembly pipeline, sensitivity analyses, and calibration checks are fully reproducible from the public code repository at https://github.com/rupumped/kidney-donor-le.

Acknowledgements

Acknowledgements

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